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Abstract

Hybrid Vision-Based Geometric Planning and Force-Aware Imitation Learning for Adaptive Robotic Surface Finishing

Robotic surface finishing requires both geometry-dependent global coverage and contact-dependent local interaction. Geometry-based planning can provide interpretable surface paths and explicit coverage constraints, but repeated changes in workpiece pose, robot initial configuration, and requested region of interest require repeated perception, registration, inverse kinematics, and motion planning. End-to-end visuomotor imitation learning can generate actions with low online inference latency, but vision-only policies do not directly observe contact pressure, slip, jamming, or weak contact, and they do not by themselves guarantee deterministic full-surface coverage.

This thesis proposes a hybrid framework that learns where interaction-dependent adaptation is required and uses geometric planning where deterministic coverage can be explicitly specified. The local component is an ROI-mask-conditioned force-aware imitation-learning policy. The policy receives visual observation, an externally generated region-of-interest mask, robot proprioception, and recent six-axis force/torque history, and generates task-conditioned pose–force action chunks. The learned reference is executed through a low-level admittance controller, which converts the error between measured wrench and desired force reference into compliant motion correction. The global component reconstructs or registers the workpiece surface, generates tool-footprint-aware coverage paths, assigns tool orientation from local surface normals, and executes the resulting trajectory with the same

contact-aware execution layer.

Two workpieces are used to evaluate the framework. One is an automotive headlight, where an ArUco marker is used only as a low-cost command interface that emulates an external defect-map request. The marker ID is converted outside the policy into a spatial ROI mask; the policy observes the mask rather than the marker ID. The local polishing phase is followed by vision-based full-coverage polishing of the headlight surface. The other is an aluminum machining specimen for pose-randomized deburring or deburring-like edge-following contact execution. This benchmark compares a classical perception–registration–planning pipeline with a vision-conditioned IL policy under variations in workpiece pose, robot initial configuration, target feature, and approach direction.

The proposed experiments evaluate ROI-mask-conditioned local processing, local-to-global hybrid polishing feasibility, online latency and cycle-time decomposition, and the role of force observation, learned force references, and admittance execution. Quantitative values for task success, force error, cycle time, and surface-quality metrics are not reported in this draft and remain [Planned/TBD: to be filled after real-robot experiments and inspection measurements].

Keywords: Robotic Surface Finishing, Geometric Planning, Coverage Path Planning, Imitation Learning, Force/Torque Sensing, ROI Mask, Admittance Control, Robotic Polishing, Robotic Deburring

Chapter 1

Introduction

1.1 Background and Motivation

Robotic surface finishing, including polishing, sanding, deburring, grinding, and buffing, is a physically demanding and quality-sensitive operation. Integrated systems such as RoboGrind show that surface scanning, defect identification, program generation, and force-controlled execution are all relevant to practical robotic surface treatment [1]. A successful finishing process depends on the nominal tool path, surface normal, tool footprint, contact force, sliding direction, dwell time, abrasive state, and robot feasibility. These factors are coupled: a path that covers a mathematical surface does not necessarily cover the surface with the physical tool, and a geometrically valid pose may still produce unstable or insufficient contact.

Geometry-based planning is therefore indispensable for many finishing tasks. Given a CAD model or a reconstructed surface, a planner can compute coverage paths, surface-normal tool orientations, boundary offsets, and robot feasibility constraints. Recent freeform-surface coverage planners explicitly consider segmentation, contact points, and manipulator constraints [2, 3]; ergodic surface-finishing methods additionally connect tool imprints and learned motion preferences to spatial coverage objectives [4]. Such methods are strong when the target region is known and the main objective is deterministic coverage. This thesis does not treat geometry-based planning as a weak baseline or as an approach that cannot perform the task.

At the same time, surface-finishing cells increasingly face repeated variation

in workpiece pose, requested region of interest (ROI), robot initial configuration, and local contact condition. In such settings, a classical pipeline must repeatedly perform perception, reconstruction or registration, target transformation, inverse kinematics, collision-aware motion planning, trajectory post-processing, and recovery. Imitation learning (IL) offers a complementary mechanism: demonstrations and planning solutions can be compressed into a policy that generates task-conditioned action chunks with low online inference latency. This is an amortized motion-generation hypothesis, not a claim that geometric planning is infeasible.

Contact-rich finishing also exposes a limitation of vision-only policies. Images and depth observations are useful for free-space approach and task semantics, but they do not directly measure contact pressure, weak contact, slip, tool jamming, or overload. Force-conditioned visuomotor policies already combine vision, proprioception, force observation, desired force references, and compliant execution for contact-rich manipulation [5, 6]. Recent policies also study global/local force control, high-frequency force/torque feedback, and force-aware reactive fusion [7, 8]. A low-level admittance controller then converts force-reference error into compliant motion correction, following established force and compliance-control principles [9–11].

Recent studies have already demonstrated force-conditioned visuomotor policies, learned force references, compliant execution, mask-conditioned manipulation policies, surface-constrained imitation learning, robotic-polishing generative policies, and hybrid learning–geometry systems [5, 12–16]. Therefore, this thesis does not claim novelty from these individual components in isolation. Its focus is narrower: an external defect request is represented as a spatial ROI mask, interaction-dependent local processing is assigned to a force-aware policy, deterministic full-surface coverage is assigned to a geometry

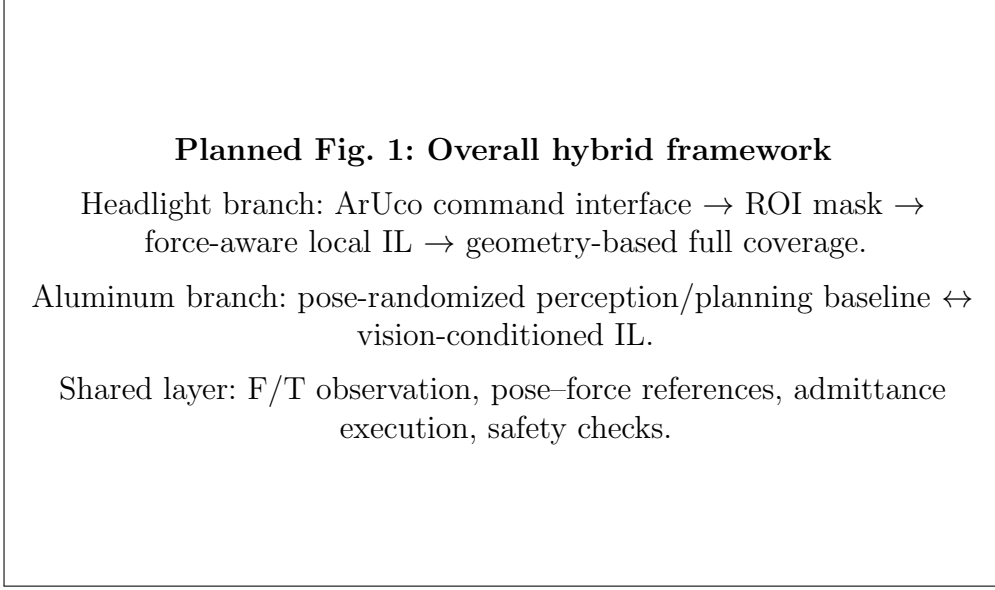


Figure 1.1: Overall structure of the proposed hybrid framework. The figure is a placeholder until the final system diagram is prepared.

planner, and the resulting workflow is evaluated with force-observation/action ablations and a planning-latency benchmark.

This thesis proposes a hybrid framework with a simple allocation principle: learn where interaction-dependent adaptation is required, and use geometric planning where deterministic coverage can be explicitly specified. The local component is an ROI-mask-conditioned force-aware IL policy that generates pose–force action chunks for contact-sensitive local processing. The global component is a vision- or reconstruction-based geometry planner that generates tool-footprint-aware full-coverage trajectories. Both components share a common execution layer based on force/torque sensing and admittance control.

1.2 Research Questions

The thesis is organized around four research questions.

RQ1: ROI-mask-conditioned local processing. Can a force-aware

imitation-learning policy generate task-appropriate local pose–force trajectories for spatially varying ROI masks on a curved workpiece? For the headlight task, the policy should respond to the location, size, and shape of the mask rather than memorize a marker ID or a fixed trajectory associated with one marker.

RQ2: Hybrid local-to-global surface finishing. Can a hybrid workflow combining force-aware IL-based local processing and geometry-based full-coverage planning provide a practical local-to-global robotic polishing framework? Before standardized physical defects are introduced, the evaluation focuses on ROI coverage, spillover, contact stability, and execution feasibility rather than physical-defect removal performance.

RQ3: Rapid response to initial-condition variations. Can a vision-conditioned IL policy achieve comparable task success with lower and more consistent online computation time than a conventional geometry-based planning pipeline under variations in workpiece pose and robot initial configuration? The comparison is between a classical perception–registration–planning pipeline and a vision-conditioned IL policy, not between “vision” and “learning” in isolation.

RQ4: Contribution of force-aware observation and action. Do six-axis force/torque observations and learned force references improve contact stability and task execution compared with pose-only IL and fixed-force execution? This question separates the effect of force observation, force action, and admittance-based execution.

1.3 Contributions

The thesis makes the following planned contributions:

1. a force-aware imitation-learning framework that uses ROI masks, visual

observation, proprioception, and force/torque history to generate local pose–force trajectories;

2. a defect-map-conditioned local polishing architecture in which ArUco markers are used only as an external command interface for ROI-mask generation and are not provided as policy observations;
3. a local-to-global hybrid surface-finishing framework that assigns state-dependent local processing to force-aware IL and deterministic full-surface coverage to vision/geometry-based planning;
4. an experimental framework that compares classical geometry-based planning and IL policy execution for pose-randomized aluminum deburring, separating task success, online latency, cycle time, and contact stability.

These contributions are stated as proposed and evaluated components. Superiority over the baselines is a hypothesis to be tested, not an assumed result.

1.4 Thesis Organization

Chapter 2 reviews robotic surface finishing, coverage planning, force control, visuomotor IL, force-aware IL, mask-conditioned policies, and learned or amortized planning. Chapter 3 defines the system, coordinate frames, workpieces, task allocation, and shared pose–force interface. Chapter 4 presents vision-based geometric surface planning for headlight coverage and the aluminum geometry-based baseline. Chapter 5 defines the ROI-mask-conditioned force-aware IL policy and admittance execution. Chapter 6 describes the hybrid local-to-global headlight workflow. Chapter 7 presents the experimental design and metrics for the headlight and aluminum studies. Chapter 8 concludes the thesis and outlines future work.

Chapter 2

Background and Related Work

2.1 Robotic Surface Finishing

Robotic polishing, grinding, sanding, and deburring combine geometric path generation with physical interaction. RoboGrind is a close system-level reference because it combines 3-D scanning, defect identification, robot program generation, and force-controlled surface treatment in an integrated workflow [1]. This thesis therefore does not treat the combination of inspection, geometric planning, and force-controlled execution as new by itself.

Geometry-based coverage planning remains the standard mechanism when a surface, tool footprint, boundary, and feasibility model are available. Freeform-surface planning studies have considered segmentation, tool geometry, contact point selection, curvature, and manipulator/task constraints [2, 3]. Ergodic surface finishing further uses a desired spatial distribution, tool imprint, and learned motion preferences to guide online surface treatment [4]. These works support the global stage of this thesis: deterministic coverage is assigned to a geometric planner rather than to an end-to-end policy.

Learning-based surface motion has also entered polishing and surface-constrained manipulation. Visual-guided Diffusion Policy with Mesh-DMP integrates learned action generation with explicit surface-constrained motion embedding for robotic freeform polishing [13]. Surface Constraint Policy uses visual observations and learned surface constraints to generate dynamically feasible surface-constrained skills [14]. Stage-Aware and Roughness-Constrained

Diffusion Policy targets multi-stage robotic polishing and generates process-related feed speed and normal contact force under roughness-oriented constraints [15]. These studies overlap strongly with surface-aware action generation and process-parameter learning; the distinction in this thesis is the explicit allocation of local ROI-conditioned interaction to force-aware IL and global full-coverage execution to a geometry planner.

Deburring is related but not identical to polishing. A rigid cutter, abrasive point, or compliant finishing tool follows an edge, hole, or chamfer while maintaining acceptable tool load and avoiding collision. Learning-guided force-feedback MPC for robotic deburring combines a diffusion motion prior, force-feedback MPC, and obstacle avoidance [17]. This thesis cites that work as a deburring and learned-warm-start reference, not as evidence for validated material-removal quality in the present aluminum experiment. If the aluminum specimen is smooth or burr geometry is not measured, the task is reported as force-controlled edge-following or deburring-like contact execution.

2.2 3-D Reconstruction and Coverage Path Planning

Coverage planning on a three-dimensional surface seeks a path whose swept tool footprint covers the target region. The important footprint is the effective contact footprint of the polishing or deburring tool, not the tool center point. Freeform-surface planners and system studies motivate the use of surface normals, tool-contact modeling, adaptive spacing, segmentation, reachability checks, and manipulator constraints [1–3, 18].

The limitation considered here is not task impossibility. Rather, when the same rigid task must be repeated under many workpiece poses, robot initial configurations, or requested ROIs, the perception–registration–IK–planning

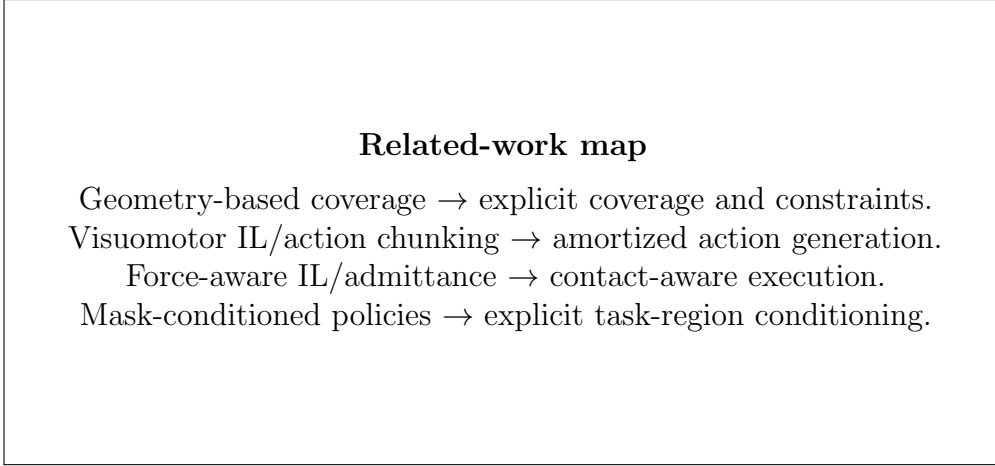


Figure 2.1: Conceptual map of the literature groups used to motivate the proposed role allocation. The figure is a placeholder.

loop is repeatedly invoked. The proposed framework keeps geometry-based planning as the deterministic coverage mechanism and as a strong classical baseline, while evaluating whether learned policies can amortize online computation for repeated local or pose-conditioned queries.

2.3 Force, Impedance, and Admittance Control

Force and compliance control are standard tools for contact-rich robotic manipulation [9–11]. In admittance control, measured force error is mapped into motion correction. This is appropriate for industrial position-controlled manipulators that accept pose or velocity commands. The high-level policy should therefore be distinguished from the low-level controller: the policy proposes task-conditioned pose and force references, while the controller stabilizes contact and enforces the reference within safety limits.

2.4 Visuomotor Imitation Learning and Generative Policies

Visuomotor imitation learning maps sensory observations to robot actions from demonstrations. Action chunking predicts a short future sequence rather than a single immediate command; ACT introduced this idea with transformer policies for fine-grained manipulation [19]. Diffusion Policy models multimodal action distributions through conditional action diffusion and receding-horizon execution [20]. Flow-matching policies are relevant when the implemented backbone uses flow-based generation: Riemannian Flow Matching Policy studies manifold-aware robot motion learning with fast inference [21].

These references are used to position the policy-learning mechanism, not to rename the implemented method. If the final implementation uses Flow Matching, Chapter 5 describes it as a conditional flow-matching action-chunk policy. ACT and Diffusion Policy remain related-work references for action chunks and generative action distributions.

2.5 Force-Aware and Compliance-Aware Imitation Learning

Recent work has added force and tactile information to imitation learning for contact-rich manipulation. Admittance Visuomotor Policy Learning is the most direct structural overlap: it uses vision, proprioception, and force observation to generate action trajectories and desired contact forces, then executes them with an admittance controller [5]. ForceMimic emphasizes force-motion capture, wrench-position parameter prediction, and hybrid force-position execution [6]. FAWAM encodes historical six-axis force/torque signals, jointly predicts future actions and wrenches, and uses force residual correction

during execution [22]. Force Policy separates a global vision policy from a high-frequency local force policy in an interaction frame for hybrid force-position control [7]. FoAR uses high-frequency F/T input with visual observations for reactive multimodal fusion [8].

Other compliance-aware policy studies are relevant to the controller boundary. Adaptive Compliance Policy learns approximate compliance profiles for diffusion-guided control [23]; Reactive Diffusion Policy combines a slow visual policy with a fast tactile/force loop [24]; Flow with the Force Field studies compliant flow-matching policies from force and demonstration-guided simulation data [25]; and Force Generative Imitation Learning estimates force commands from position trajectories through a control-oriented formulation [26]. These works mean that F/T observation, F/T history, pose-plus-force output, desired force references, admittance execution, high-frequency force correction, and polishing force generation are not treated as standalone novelty claims in this thesis.

2.6 Mask-Conditioned Policies

Mask-conditioned policies provide task-relevant spatial information explicitly. Transferring Foundation Models for Generalizable Robotic Manipulation uses raw RGB and segmentation masks in a two-stream imitation-learning policy, making the semantic mask an explicit task condition [12]. VIOLA uses object proposal priors to form object-centric representations and attend to task-relevant visual factors [27]. Mask2Act uses future object masks as a latent plan for action-free video pre-training [28]. GROOT uses object-centric 3-D representations and segmentation correspondence for generalizable manipulation [29].

The proposed ROI mask is therefore not presented as a new visual representation by itself. It is used as an explicit and transferable task-specification interface for local force-aware surface processing. In the headlight setup, the mask is generated from an external command or inspection request. The initial ArUco implementation is only a low-cost command interface [30]; the marker ID is not a policy observation and is not treated as visual evidence of a physical defect.

2.7 Learned and Amortized Motion Planning

Learning-based motion planning can be interpreted as amortizing repeated planning queries. Motion Planning Networks learn fast motion-plan predictions and can be combined with classical planning for feasibility repair [31]. MPC-MPNet combines learned proposals with model-predictive refinement under kinodynamic constraints [32]. Motion Planning Diffusion learns trajectory-distribution priors with diffusion models and adapts them with motion-planning costs [33]. Point-cloud-conditioned flow matching has also been studied as a warm start for optimization-based manipulator motion planning [34].

This perspective motivates RQ3. The aluminum experiment does not claim that IL generally replaces classical planning. Classical planners retain advantages in collision checking, explicit constraints, and feasibility repair. The planned comparison measures latency, latency variance, cycle time, contact stability, and task success under a shared execution layer.

2.8 Related-Work Comparison

Table 2.1: Related-work comparison. Y = reported or central to the method, N = not part of the method, N/R = not reported or not verified from the checked source. The table is intended to bound novelty rather than rank methods.

Work	Task/application	RGB/RGB-D observation	ROI/object mask condition	F/T observation	Force-history input	Pose/action prediction	Force/surface prediction	Compliance/admittance co-actuation	Explicit surface geometry	Explicit global coverage	Local/global decomposition	Defect-map interface	Planning latency evaluation	Surface-quality evaluation
Adaptive Visionmotor Policy [1]	General contact-rich manipulation	Y	N	Y	N/R	Y	Y	Y	N	N	N	N	N/R	N
Prochaine [6]	Force-aware contact-rich manipulation	N/R	N	Y	N/R	Y	Y	Y	N	N	N	N	N/R	N
Force Policy [7]	Force-aware contact-rich manipulation	Y	N/R	Y	Y	Y	Y	Y	N	N	Y	N	N/R	N
FAWAM [22]	Closed-loop manipulation	N/R	N/R	Y	Y	Y	Y	Correction	N	N	N	N	N/R	N
FeAR [6]	Reactive contact-rich manipulation	Y	N	Y	Y	Y	N/R	Position/reactive	N	N	N	N	N/R	N
Adaptive Compliance Policy [23]	Compliance-aware manipulation	Y	N/R	N/R	N/R	Y	N/R	Compliance	N	N	N	N	N/R	N
Force-aware Diffusion Policy [24]	Force-aware contact manipulation	Y	N/R	Twists/force	Y	Y	N/R	Reactive loop	N	N	Y	N	N/R	N
Flow with the Force [25]	Compliant manipulation	Y	N/R	Y	N/R	Y	N/R	Compliance	N	N	N	N	N/R	N
Multi-stage Electric SRP [15]	Overmatching policy manipulation	Y	N/R	N/R	Y	Y	Normal force	Process constraint	Surface task	N/R	Stages-based	N	N/R	Y
SCT [14]	Surface-constrained polishing	Y	N	N/R	N/R	Y	N/R	Compliant DMP	Y	N	N	N	N/R	N/R
Visual-Guided Diffusion Policy [13]	Free-form surface manipulation	RGB-D	N	N/R	N/R	Y	N/R	N/R	Y	N/R	N	N	N/R	N/R
+ Mesh-DMP [13]	Free-form surface manipulation	Y	Y	N	N	Y	N	N/R	N	N	Local-global visual	N	N/R	N
RobotGrind [1]	Industrial surface finishing	3-D scan	Defect region	Y	N/R	Program/path	Force setpoint	Hybrid force-position	Y	Path planning	Workflow stages	Y	N/R	Y
Ergodic Surface Finishing [1]	Surface finishing with learned preferences	N/R	Spatial distribution	N/R	N/R	Median preference	N/R	N/R	Y	Y	N/R	N/R	N/R	N/R
Proposed method	Force-aware polishing and pose-randomised deburring	Y	Y	Y	Y	Y	Verified force channel	Y	Y	Y	Y	Y	Planned	Planned

The comparison shows that most individual components of the proposed workflow already exist in the literature. The thesis therefore positions its contribution as a task-specific integration and evaluation: an external ROI interface for local contact-rich surface processing, force-aware local pose–force generation, deterministic full-surface coverage, and a latency/cycle-time benchmark against a classical perception–registration–planning pipeline. This is not framed as an absolute novelty claim.

2.9 Research Gap

The reviewed literature motivates the thesis around a specific intersection. Surface-finishing systems already integrate perception, planning, and force control; force-aware IL already predicts motion and force-related references; mask-conditioned manipulation already uses spatial task conditions; and learned planners already amortize repeated planning queries. The planned contribution is to evaluate a local-to-global surface-finishing workflow in which interaction-dependent local processing and deterministic global coverage are assigned to different solvers, and to test the resulting force-aware ablations and planning-latency trade-offs on complementary headlight and aluminum workpieces.

Chapter 3

System Overview and Problem Formulation

3.1 System Components

The experimental platform is organized around a six-degree-of-freedom industrial manipulator, a finishing spindle or deburring tool, a wrist-mounted six-axis force/torque sensor, an RGB-D sensor such as an Intel RealSense D435 for surface reconstruction or registration, a local RGB or RGB-D camera for policy observation, and rigid fixtures for the headlight and aluminum workpieces. The exact robot model, controller rate, sensor mounting, tool type, and calibration procedure remain [Planned/TBD: to be finalized from the real hardware setup].

The system has three high-level modules. The geometry-based planner reconstructs or registers the workpiece, computes surface normals and feasible tool poses, and generates deterministic coverage or baseline trajectories. The force-aware IL policy generates local task-conditioned pose-force references from sensory observations, a structure that directly overlaps with recent admittance visuomotor and hybrid force-position policies [5, 7]. The low-level admittance controller executes the references while enforcing force, velocity, workspace, and collision limits.

3.2 Coordinate Frames

Let $\{B\}$ denote the robot base frame, $\{E\}$ the end-effector frame, $\{T\}$ the tool frame, $\{F\}$ the force/torque sensor frame, $\{C_g\}$ the global RGB-D

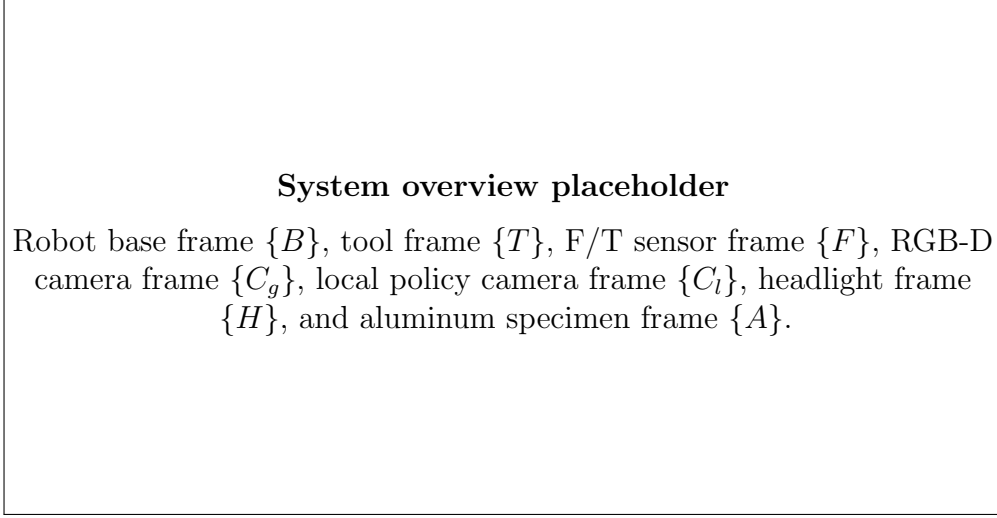


Figure 3.1: Coordinate frames and shared modules of the proposed system. The final figure should show the actual robot, cameras, workpieces, and calibration transforms.

camera frame, and $\{C_l\}$ the local policy camera frame. The headlight and aluminum workpiece frames are denoted by $\{H\}$ and $\{A\}$, respectively. A calibrated rigid transform is written as ${}^i\mathbf{T}_j \in SE(3)$ from frame $\{j\}$ to frame $\{i\}$.

The global geometric pipeline estimates ${}^B\mathbf{T}_H$ or ${}^B\mathbf{T}_A$ from scan-to-CAD registration, fiducial-assisted calibration, or fixture calibration. The IL policy should not receive marker IDs or global registration outputs as a shortcut unless that information is explicitly part of the intended deployed observation. For the headlight local task, the policy receives a spatial ROI mask in the local camera or headlight coordinate system. ArUco is cited only as a fiducial detection and command-interface mechanism, not as evidence of physical surface defects [30].

3.3 Workpieces

Two workpieces are considered.

1. **Automotive headlight:** a curved transparent or coated surface used for

local ROI-mask-conditioned polishing followed by full-surface geometry-based polishing. The current thesis treats ArUco markers as a command interface for ROI generation, not as defect-inspection sensors or defect evidence [30].

2. **Aluminum machining specimen:** a rigid part containing holes, edges, or deburring-like target features. The benchmark varies workpiece pose, robot initial configuration, target feature, and approach direction to compare a classical online planning pipeline with a vision-conditioned IL policy.

3.4 Hybrid Task Allocation

Let $\mathcal{S}$ be a workpiece surface and $\Omega \subset \mathcal{S}$ be the target finishing region. The task is divided into a deterministic global region Ω_g and local requested regions Ω_ℓ^i :

$$\Omega = \Omega_g \cup \bigcup_{i=1}^{N_\ell} \Omega_\ell^i. \quad (3.1)$$

The global region is assigned to geometry-based planning when coverage completeness, boundary handling, and tool footprint are explicitly defined. A local region is assigned to IL when the required behavior depends on ROI mask shape, local curvature, approach direction, contact onset, force modulation, or a changing initial condition. Surface-constrained policy learning already addresses geometric consistency for learned surface actions [14]; this thesis instead separates local learned interaction from explicit global coverage. This allocation does not imply that geometry-based planning is unable to process local regions; instead, the thesis evaluates whether IL can provide lower and more consistent online computation while preserving task quality.

3.5 Common Pose–Force Interface

Both the geometric planner and IL policy output references for the same low-level execution layer. A high-level action chunk is written as

$$a_{t:t+H} = [X_{t:t+H}^d, W_{t:t+H}^d], \quad (3.2)$$

where X^d is a future end-effector pose or pose-increment trajectory and W^d is the desired force or wrench reference. The current repository does not contain the robot dataset or controller source code needed to verify whether the implemented action is Cartesian pose, Cartesian increment, joint-space command, normal-force-only reference, or multi-axis wrench. This draft therefore states the intended interface and leaves the exact channel definition as [Planned/TBD: verify against the actual dataset and controller implementation].

If the actual implementation commands only normal force F_z^d , the action should be written as $a_{t:t+H} = [X_{t:t+H}^d, F_{z,t:t+H}^d]$ rather than as a full six-dimensional wrench. If the implementation commands three force components, the action should be written as $[X^d, F_x^d, F_y^d, F_z^d]$. The text avoids claiming six-axis wrench control until the controller channels are verified.

3.6 Admittance Execution

The low-level controller maps force-reference error into motion correction, following classical impedance/admittance-control principles and recent learned desired-force execution structures [5, 9–11]:

$$\mathbf{M}_d \Delta \ddot{\mathbf{x}} + \mathbf{D}_d \Delta \dot{\mathbf{x}} + \mathbf{K}_d \Delta \mathbf{x} = \mathbf{W}_{\text{ext}} - \mathbf{W}^d. \quad (3.3)$$

Here $\Delta \mathbf{x}$ is the compliant Cartesian correction, $\mathbf{W}_{\text{ext}}$ is the measured external wrench or selected force channel, and $\mathbf{W}^d$ is the desired reference from the planner or policy. The signs and gains must be matched to the deployed

controller. The conceptual role is fixed: the policy generates task-conditioned references, the F/T sensor observes contact state, and admittance control performs stable compliant execution.

3.7 Problem Definitions

The headlight task is to process a requested local ROI and then execute full-coverage polishing over the visible target surface. Before standardized defect specimens are introduced, success is defined through ROI tool-footprint coverage, spillover outside the ROI, contact stability, completion time, and feasibility of the local-to-global sequence.

The aluminum task is to execute a specified edge-following or deburring-like contact trajectory under randomized workpiece pose and robot initial configuration. If real burrs are not present or not measured, the task is reported as force-controlled edge-following or deburring-like contact execution rather than a validated material-removal deburring study. The main hypothesis is that the IL policy can provide comparable task success with lower and more consistent online computation time than the classical pipeline under the same safety and execution limits.

Chapter 4

Vision-Based Geometric Surface Planning

4.1 Motivation and Scope

Geometry-based planning remains the preferred mechanism when the task can be specified through surface geometry, tool footprint, boundaries, and robot constraints. Freeform-surface finishing planners motivate explicit use of segmentation, contact geometry, coverage checks, and manipulator constraints [2,3]. This chapter defines the deterministic full-coverage planner for the headlight surface and the classical perception–registration–planning baseline for the aluminum specimen. The purpose is to establish a strong classical component, not a deliberately slow or incomplete baseline.

4.2 Headlight Reconstruction

The headlight global-polishing branch starts from an RGB-D scan, a calibrated fixture pose, or a CAD/scan registration. Similar scanning-to-program workflows appear in robotic surface-treatment systems such as RoboGrind [1]. The intended pipeline is:

1. acquire the headlight surface with the global RGB-D sensor;
2. remove background, fixture, and marker regions that are not part of the polishing surface;
3. estimate or register the surface in the robot base frame;

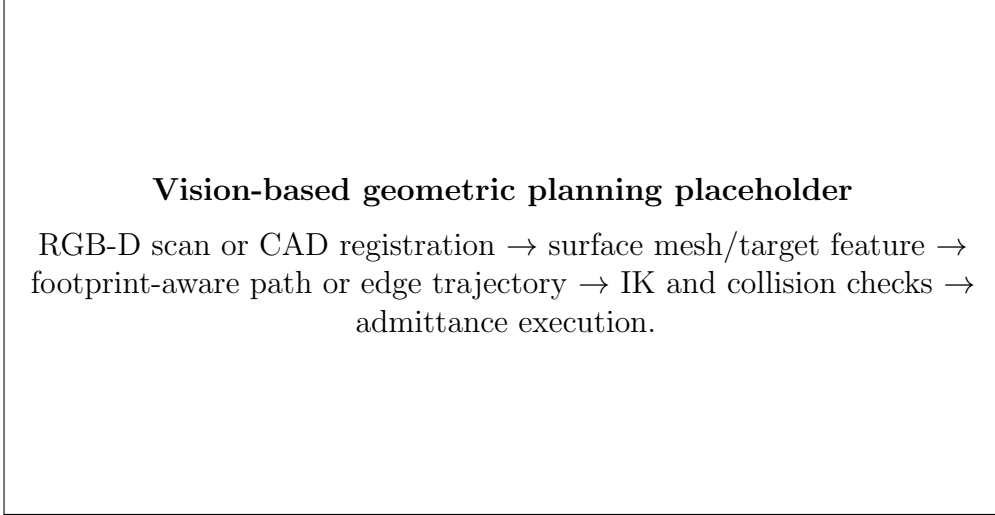


Figure 4.1: Planned geometric planning pipeline for full-coverage polishing and the aluminum classical baseline. The figure is a placeholder.

4. reconstruct a mesh or filtered point cloud with normals and boundaries;
5. define the global target surface excluding protected areas and local ROI-handled regions.

If a D435 is used, it is treated as a 3-D surface sensor for reconstruction or registration. It is not described as a defect-inspection sensor. Real defect maps require a separate high-resolution inspection system and are left as future work unless such a system is implemented.

4.3 Tool-Footprint-Aware Coverage

Let $\gamma(s) \in \mathcal{S}$ be the centerline of a polishing path on the reconstructed or registered surface, and let $\mathcal{B}(\gamma(s), \mathbf{R}(s), w_{\text{eff}})$ be the surface footprint swept by the polishing tool at path parameter s . The full-coverage objective is

$$\Omega_g \subseteq \bigcup_{s \in [0, L]} \mathcal{B}(\gamma(s), \mathbf{R}(s), w_{\text{eff}}), \quad (4.1)$$

up to a documented tolerance near protected boundaries, occluded regions, and physically inaccessible areas. The effective footprint w_{eff} depends on pad

diameter, compression, local curvature, tool tilt, and the selected contact point. It is not equivalent to TCP path spacing.

For a desired overlap ratio η , the nominal cross-track spacing is

$$d(\mathbf{q}) = (1 - \eta)w_{\text{eff}}(\mathbf{q}), \quad (4.2)$$

with local reductions in high-curvature or uncertain regions. Candidate path directions can be generated from raster projection, geodesic offsets, or surface parameterization and then selected according to coverage, path length, overlap, boundary behavior, and robot feasibility. Ergodic surface finishing is related because it uses tool-imprint and spatial-distribution objectives for surface treatment [4]; the present planner is limited to a deterministic implementation of the global coverage stage after local ROI processing.

4.4 Orientation Generation and Robot Feasibility

Tool orientation is initialized from the local surface normal $\mathbf{n}(\mathbf{q})$ and path tangent $\mathbf{t}(\mathbf{q})$. Orientation smoothing limits angular acceleration while preserving the process-specific tilt range. Each waypoint is checked for inverse kinematics, joint limits, singularity margin, self-collision, fixture collision, workpiece collision, and tool-body clearance. Infeasible segments are either replanned, assigned to a safer contact point, or excluded with a documented coverage gap. Learned surface-action approaches such as Mesh-DMP-integrated diffusion policy and Surface Constraint Policy also address geometric consistency of learned actions [13, 14]; this chapter keeps that role in an explicit geometric planner for the global phase.

4.5 Full-Coverage Execution

The global coverage trajectory is converted to the common pose–force interface of Eq. (3.2). The pose sequence follows the reconstructed or registered surface. The force reference is a conventional fixed normal-force schedule or a process-specific profile unless a verified force-action policy is used. Execution uses the same admittance controller as the learned local policy so that the comparison does not confound planning strategy with low-level contact control.

4.6 Aluminum Geometry-Based Baseline

The aluminum benchmark uses a classical pipeline:

1. acquire a D435 scan or the actual perception input used in the system;
2. register the observed part to a nominal CAD or mesh model;
3. estimate the workpiece pose and target edge or hole transform;
4. compute IK for the approach, processing, and retraction poses;
5. perform collision-aware motion planning from the current robot initial configuration;
6. generate a parameterized deburring or edge-following trajectory;
7. execute the trajectory with the conventional force/admittance layer.

The baseline records sensor acquisition time, reconstruction time, registration time, IK time, motion-planning time, trajectory post-processing time, execution time, and recovery time. It should be implemented as a strong classical method. A slow script or unnecessarily repeated computation is not an acceptable baseline for claiming reduced IL latency. Learned motion-planning references are used only to frame amortized online computation, not to weaken the classical baseline [31, 33].

4.7 Classical Local ROI Baseline

For the headlight study, a geometry-derived local baseline is required in addition to full-coverage-only polishing. Given an ROI mask, this baseline generates a local raster or contour path within the mask, assigns tool orientation from the local surface normal, and executes with fixed force or conventional force control. This baseline addresses the question of whether a simple ROI-conditioned raster path is sufficient. The proposed force-aware IL method should be evaluated against this baseline before any claim about the value of learned local processing is made.

4.8 Relation to Process Optimization

Previous drafts of this thesis included reinforcement learning for Preston-rate-based sliding-velocity optimization. That topic is not treated as a main contribution in the present thesis direction. The Preston relation remains useful background for understanding material removal, but no RL velocity result is used as evidence for the proposed hybrid framework. Future work may revisit speed or process-parameter optimization after the geometry/IL role allocation and force-aware contact execution have been validated.

Chapter 5

ROI-Mask-Conditioned Force-Aware Imitation Learning

5.1 Motivation and Scope

The local policy is designed for short contact-rich tasks whose desired action depends on a requested ROI, local curvature, initial robot state, and contact history. In the headlight task, the ROI mask specifies the area to process before global polishing. In the aluminum task, vision and state observations specify the current pose-conditioned approach and contact execution. The policy is not intended to replace geometric planning for broad surfaces where explicit coverage can be verified.

Imitation learning is used because expert demonstrations can encode approach, contact establishment, local traversal, force modulation, and retraction without hand-designing a new rule set for every ROI and initial condition. The thesis treats IL as amortized motion generation: online inference should be low and consistent after offline demonstration collection and training.

Three related-work groups bound the scope of this chapter. First, force-conditioned action generation has already been studied through admittance visuomotor policies, ForceMimic, FAWAM, Force Policy, FoAR, compliant flow matching, and stage-aware robotic polishing policies [5–8, 15, 22, 25]. Second, compliant execution and contact correction are grounded in classical admittance/impedance control and have been combined with learned policies in recent contact-rich manipulation work [9–11, 23, 24]. Third, spatial semantic conditioning through masks or object-centric representations has been used

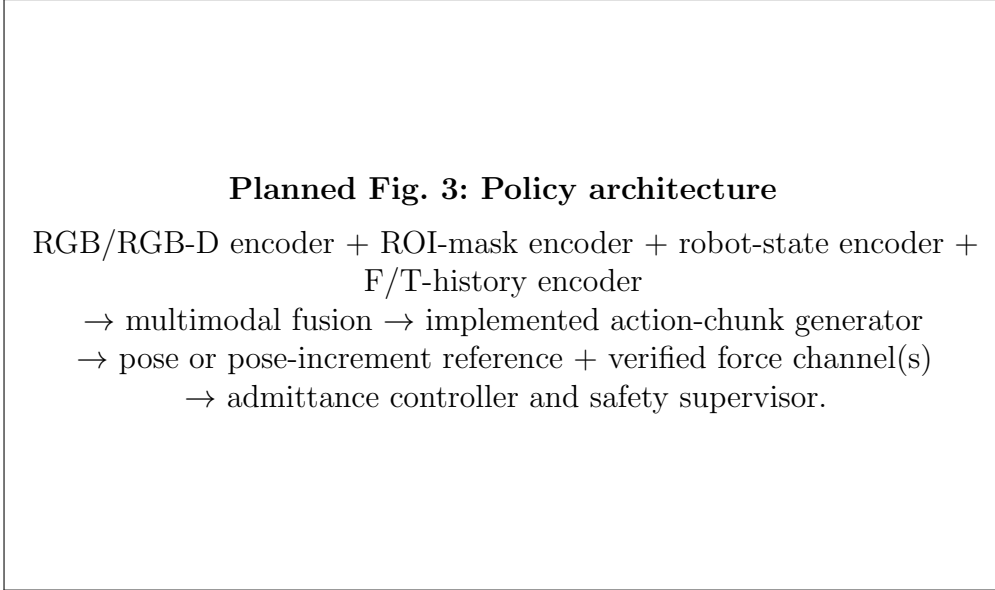


Figure 5.1: ROI-mask-conditioned force-aware IL architecture. The final figure should reflect the implemented backbone and the actual action channels.

in manipulation policies [12, 27–29]. The proposed policy does not claim novelty from force sensing, force-reference prediction, admittance execution, action chunking, generative policy learning, or mask conditioning in isolation. It investigates an external ROI-mask task interface for local contact-rich surface processing and integrates it with deterministic global coverage and a planning-latency benchmark.

5.2 Demonstration Data

Each demonstration episode should contain synchronized local RGB or RGB-D observations, ROI masks when applicable, robot state, end-effector pose, measured six-axis force/torque, commanded force reference if available, spindle or tool state, phase labels, intervention labels, and success or failure annotations. Demonstrations may be collected through teleoperation, kinesthetic teaching, scripted baseline execution corrected by an expert, or a force-motion capture system such as the type studied in ForceMimic [6].

Force signals are bias-corrected, transformed to a consistent frame, synchronized with images and robot state, and filtered without removing contact onset or overload transients. Failed demonstrations should be rejected or labeled explicitly. The dataset must include continuous ROI variation rather than only a few marker IDs connected to fixed trajectories.

5.3 Observation and Action

For ROI-conditioned local processing, the intended observation is

$$o_t = [I_t, M_{\text{ROI},t}, s_t, W_{t-h:t}], \quad (5.1)$$

where I_t is the local RGB or RGB-D observation, $M_{\text{ROI},t}$ is a binary or soft ROI mask, s_t is the robot proprioceptive state, and $W_{t-h:t}$ is a recent history of measured six-axis force/torque. For the aluminum benchmark, the ROI mask term may be omitted or replaced by a target-feature mask if such a mask is used. Force-history-conditioned prediction is compared with FAWAM and FoAR, which explicitly use recent or high-frequency force/torque signals for closed-loop contact-rich manipulation [8, 22]. The ROI mask stream is compared with segmentation- and object-mask-conditioned manipulation policies [12, 27, 28].

The intended action is an H -step chunk:

$$a_{t:t+H} = [X_{t:t+H}^d, W_{t:t+H}^d]. \quad (5.2)$$

Here X^d denotes future end-effector pose or pose increments, and W^d denotes the desired force or wrench reference. Desired force output is compared with Admittance Visuomotor Policy Learning, ForceMimic, FAWAM, and SRDP [5, 6, 15, 22]. The repository does not include the final dataset schema, so the exact action dimension is [Planned/TBD: to be verified]. If only normal force is predicted, the method will be reported as pose plus desired normal-force trajectory, not as six-axis wrench prediction.

5.4 Policy Architecture

The policy contains four encoders. A visual encoder extracts features from I_t . A mask encoder processes $M_{\text{ROI},t}$ and may use masked pooling or concatenation with image features. A state encoder processes robot configuration, end-effector pose, velocity, and phase variables when available. A temporal force encoder processes $W_{t-h:t}$ with a temporal convolution, recurrent model, or transformer block.

The fused representation conditions the implemented action-chunk generator. Action chunking is related to ACT [19]; generative visuomotor action distributions are related to Diffusion Policy [20]; and flow-matching robot policies are related when the implementation uses a flow-based backbone [21]. The thesis does not rename the implemented backbone. If the working implementation uses Flow Matching, the policy should be described as a conditional flow-matching action-chunk policy. ACT and Diffusion Policy remain related-work references, not substitutes for the implemented method.

5.5 Training Objective

For deterministic behavior cloning, the training loss can be written as

$$\mathcal{L}_{\text{BC}} = \lambda_X \left\| X_{t:t+H}^d - \hat{X}_{t:t+H}^d \right\|_2^2 + \lambda_W \left\| W_{t:t+H}^d - \hat{W}_{t:t+H}^d \right\|_2^2 + \lambda_{\Delta} \mathcal{L}_{\text{smooth}}, \quad (5.3)$$

where hats denote policy predictions. If a flow-matching or diffusion model is implemented, $\mathcal{L}_{\text{BC}}$ is replaced by the corresponding conditional generative objective while preserving the same observation and action semantics. Training/validation splits must include held-out combinations of mask location, size, orientation, curvature region, and workpiece pose perturbation.

5.6 Inference and Execution

During inference, the policy observes the current image, mask, state, and force history and predicts a future chunk. A receding-horizon executor sends only the initial portion of the chunk to the admittance controller before requesting a new prediction. This is an observation-conditioned response with closed-loop force correction, not online parameter learning.

5.7 Force Reference and Admittance Control

The policy and controller play separate roles:

- the IL policy generates task-conditioned pose and force references;
- the F/T sensor provides contact-state observations and feedback;
- the admittance controller converts force-reference error into compliant motion correction;
- the safety supervisor limits force, velocity, workspace, and collision risk.

The same Eq. (3.3) is used conceptually for both local IL execution and geometry-based trajectories. This policy–controller separation is directly compared with learned desired-force execution in Admittance Visuomotor Policy Learning and hybrid force-position execution in ForceMimic and Force Policy [5–7]. Any final thesis version must update signs, selected channels, and gain matrices to match the deployed controller.

5.8 Ablation Definitions

The force-aware study separates four policy variants:

1. **A1 Pose-only IL:** vision and robot state observation, pose trajectory action, and position control or fixed-force execution.

2. **A2 Force-observation IL:** vision, robot state, and F/T history observation, but pose trajectory action only.
3. **A3 Force-action IL:** vision and robot state observation with pose plus desired force trajectory output.
4. **A4 Proposed force-aware IL:** vision, ROI mask when applicable, robot state, and F/T history observation with pose plus desired force trajectory output executed by admittance control.

The evaluation compares success rate, force tracking error, peak force, force variance, contact-loss count, task time, ROI or edge-following quality, and held-out-condition performance.

Chapter 6

Hybrid Local-to-Global Surface Finishing

6.1 Motivation

The headlight demonstration combines two task types. A local ROI requires condition-dependent contact behavior: approach, contact establishment, short traversal, force modulation, and retraction. The remaining visible surface requires deterministic full-coverage polishing. The proposed workflow therefore uses force-aware IL for the requested local region and geometry-based planning for the global surface.

Several recent systems are close to this learning–geometry integration. Visual-guided Diffusion Policy with Mesh-DMP combines learned action generation with an explicit surface-constrained motion representation for freeform polishing [13]. RoboGrind integrates scanning, defect identification, robot program generation, and force-controlled surface treatment [1]. Ergodic surface finishing uses tool imprints, desired spatial distributions, and learned motion preferences for surface treatment [4]. SRDP studies stage-aware robotic polishing with feed-speed and normal-force generation under roughness-oriented constraints [15]. Learning-augmented robotic automation further frames learned controllers inside conventional industrial workflows [16]. The contribution here is therefore limited to the planned evaluation of ROI-mask-conditioned local force-aware processing, explicit global coverage, and latency/ablation comparisons on the selected workpieces.

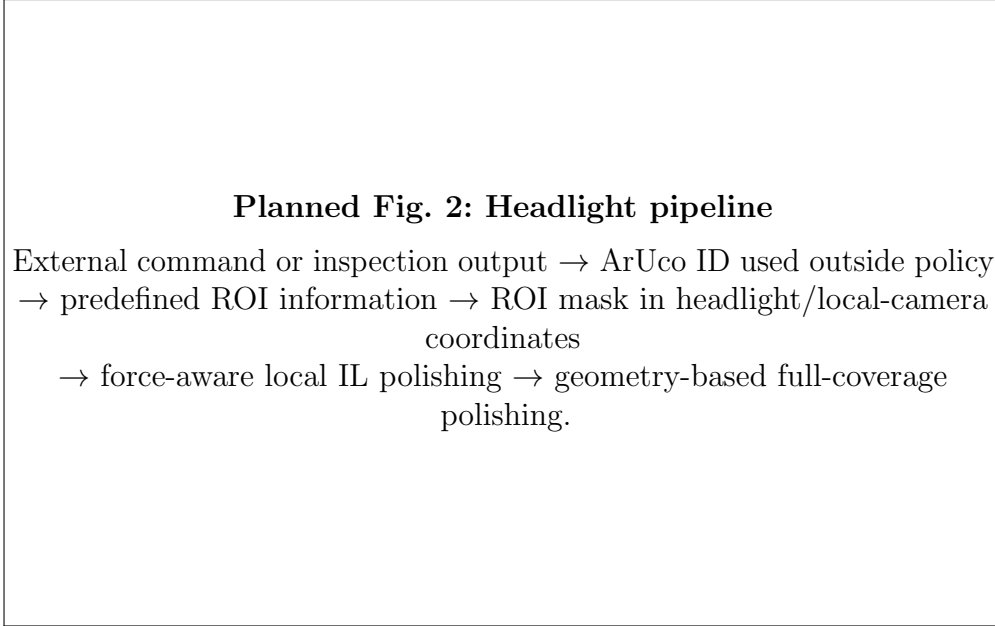


Figure 6.1: Headlight local-to-global workflow. ArUco is a command interface for ROI-mask generation, not a defect-inspection sensor or a policy observation.

6.2 ArUco-to-ROI-Mask Interface

ArUco markers are used as a low-cost command interface that emulates an external defect-request or defect-map system [30]. The marker does not sense a defect, is not treated as defect evidence, and is not polished away. The mapping is:

1. detect the ArUco marker in a command camera or a region separated from the policy image;
2. decode the marker ID outside the policy;
3. retrieve predefined ROI information or a synthetic defect-map request;
4. generate a spatial ROI mask in the headlight or local-camera coordinate frame;
5. provide the ROI mask, not the marker ID, to the IL policy.

If the marker appears in the RGB image used by the policy, the marker region

must be cropped, masked out, or captured by a separate command camera. This prevents the policy from learning a shortcut from marker pattern to trajectory. The demonstration should not be structured as a lookup table from three or four IDs to three or four trajectories.

6.3 ROI Mask Construction

The ROI mask can be binary or soft:

$$M_{\text{ROI}} : \Pi(\mathcal{S}_H) \rightarrow [0, 1], \quad (6.1)$$

where $\Pi(\mathcal{S}_H)$ denotes the image or surface projection of the headlight surface. Training masks should vary in center location, size, shape, orientation, and curvature region. Shapes may include circular, elliptical, elongated, and irregular regions. The split should include seen mask shape with unseen location, seen location range with unseen orientation, unseen size within the training range, held-out combinations of location/size/orientation, and optionally small workpiece-pose perturbations.

6.4 Local Force-Aware IL Phase

The local policy receives the observation in Eq. (5.1). It generates an action chunk for:

1. approach to a safe pre-contact pose;
2. contact establishment with bounded force overshoot;
3. local traversal over the requested ROI;
4. force modulation or force-reference tracking;
5. retract to a transition pose for global polishing.

The local success criteria are defined by footprint coverage of the ROI, spillover outside the ROI, force stability, contact-loss count, and completion time. Actual defect removal, transparency recovery, or gloss restoration is evaluated only after standardized physical defects and inspection measurements are implemented.

6.5 Global Full-Coverage Phase

After local IL inference and execution, the system executes a global path:

1. acquire or update the headlight surface reconstruction;
2. register the surface to the robot base frame;
3. process the mesh or point cloud and estimate normals;
4. generate a tool-footprint-aware coverage path;
5. assign local surface-normal-based tool orientations;
6. execute the trajectory with force/admittance control.

The local ROI phase and global phase can overlap spatially if the full-coverage process intentionally blends the local region into the rest of the surface. Any overlap is reported as part of the coverage and spillover metrics rather than treated as unmeasured physical-defect removal.

6.6 Transition, Safety, and Failure Handling

The transition from local to global processing requires a safe retract pose, a validated current robot state, and either a known workpiece transform or a fresh registration. The safety supervisor monitors workspace limits, collision margin, normal force, tangential force, tool moment, velocity, and spindle state. Failure handling includes controlled retraction, request for re-registration, rejection of

unreachable ROI requests, and operator intervention for out-of-distribution masks.

6.7 Headlight Method Variants

The headlight comparison includes four minimum variants:

1. **H1 Geometry-based full coverage only:** geometry-based global coverage without a local ROI phase.
2. **H2 Geometry-derived local + global hybrid:** ROI-mask-derived raster or contour local path with fixed force or conventional force control, followed by global coverage.
3. **H3 Pose-only IL + global hybrid:** RGB/mask/state policy that outputs pose trajectory and uses fixed-force or conventional force control, followed by global coverage.
4. **H4 Proposed force-aware IL:** RGB/mask/state/F/T-history policy that outputs pose plus verified desired force channels and executes with admittance, followed by global coverage.

H2 is essential because it tests whether a simple local geometry-derived path generated from the same ROI mask is sufficient.

6.8 Future Inspection Interface

The ArUco-based command interface is a temporary substitute for a real defect-inspection output. Future work should replace it with a high-resolution inspection system that produces a defect map or condition mask. At that stage, additional metrics such as defect-area reduction, gloss or transparency restoration, required number of passes, over-polishing area, and total cycle time

to target quality can be reported. Until then, the thesis uses the terms ROI-mask-conditioned local processing or defect-map-conditioned local processing and avoids terminology that implies autonomous defect inspection and repair.

Chapter 7

Experimental Design

7.1 Validation Strategy

The experiments evaluate the role allocation proposed in this thesis: geometry-based planning for deterministic coverage, force-aware IL for ROI- or pose-conditioned local interaction, and admittance control for contact execution. The ablation structure is motivated by recent force-aware and compliance-aware policies that separately use force observations, force-history inputs, force or wrench predictions, and hybrid/admittance execution [5–8, 22]. No numerical result is reported until the corresponding real-robot experiment or inspection measurement is completed.

7.2 Hardware and Software Setup

The hardware setup includes a six-degree-of-freedom robot, a six-axis F/T sensor, a local camera for policy observation, a global RGB-D sensor for reconstruction or registration, polishing and deburring tools, a headlight fixture, and an aluminum specimen fixture. The software setup includes the robot controller, perception and registration stack, motion-planning stack, IL inference stack, data logger, and safety supervisor. Exact model names, controller rates, synchronization method, tool geometry, and safety thresholds remain [Planned/TBD: to be filled from the final implementation].

Planned Fig. 4: Aluminum benchmark

Three to five anchor workpiece poses with continuous perturbations in translation, orientation, robot initial configuration, target feature, and approach direction.

Classical pipeline: perception $\rightarrow$ registration $\rightarrow$ IK/planning $\rightarrow$ execution.

IL pipeline: observation $\rightarrow$ policy inference $\rightarrow$ execution.

Figure 7.1: Aluminum pose-randomized benchmark. The figure is a placeholder until the final experimental layout is photographed or drawn.

7.3 Headlight ROI-Mask Experiment

The headlight experiment evaluates RQ1 and RQ2. Training data include ROI masks with varying center location, size, shape, orientation, and curvature region. The test split includes seen mask shape with unseen location, seen location range with unseen orientation, unseen mask size within the training range, held-out location/size/orientation combinations, and optional small workpiece-pose perturbation. The comparison is positioned relative to defect-to-program workflows, learning-mesh hybrids, ergodic surface finishing, stage-aware polishing policies, and mask-conditioned manipulation policies [1, 4, 12, 13, 15].

The evaluation metrics before physical defects are introduced are ROI tool-footprint coverage ratio, spillover or unnecessary processing outside the ROI, local task completion time, policy inference time, normal-force RMSE, force variance, peak force and overshoot, contact-loss count, tool moment or orientation deviation, and held-out ROI success rate. If standardized physical defects are later added, the additional metrics are defect-area reduction, gloss

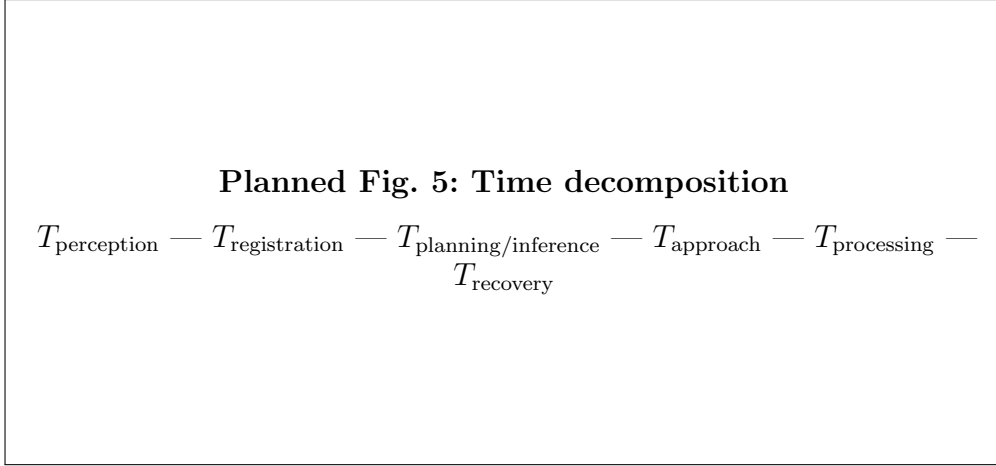


Figure 7.2: Planned visualization for cycle-time decomposition. The final figure should plot distributions rather than only means.

or transparency restoration, required number of passes, over-polishing area, and total cycle time to target quality.

7.4 Hybrid Headlight Demo

The local-to-global demonstration executes local IL or a local baseline before full-coverage geometric polishing. The minimum method variants are H1–H4 from Chapter 6. The H2 geometry-derived local + global baseline is required to test whether ROI-mask-derived local raster or contour planning is already sufficient.

7.5 Aluminum Pose-Randomized Deburring Benchmark

The aluminum benchmark evaluates RQ3. Three to five anchor poses are used only as centers of continuous randomization. Around each anchor, translation, yaw/pitch/roll, robot initial joint configuration, target hole or target edge, approach direction, and optional fixture or obstacle condition are perturbed within safe ranges. The perturbation values remain [Planned/TBD: to be set after workspace and safety validation]. The comparison is motivated

by learned and hybrid planning work that amortizes repeated queries or warm-starts classical optimization [17, 31–34].

The split is:

1. **ID test:** new combinations inside the training randomization range;
2. **Interpolation test:** poses between training anchors;
3. **Boundary test:** conditions near the edge of the training range;
4. **Mild OOD test:** conditions slightly beyond the training range;
5. **Failure-awareness test:** unreachable or unsafe target requests.

The number of trials per method and split is [Planned/TBD: TBD].

7.6 Geometry-Based Planning Versus IL

The classical baseline uses the pipeline in Chapter 4. The IL method generates approach and processing trajectories directly from the current observation, state, and force history when applicable. Both methods use the same robot, tool, velocity and acceleration limits, force limits, and low-level execution layer to isolate the effect of online planning versus policy inference.

Two timing comparisons are reported separately. The algorithm-level comparison uses matched target and pose information and compares classical planning computation time against policy inference and post-processing time. The end-to-end comparison measures the full pipeline from sensing to task completion. The thesis does not compare IL inference-only time against a classical end-to-end time.

The cycle time is decomposed as

$$T_{\text{cycle}} = T_{\text{perception}} + T_{\text{registration}} + T_{\text{planning/inference}} + T_{\text{approach}} + T_{\text{processing}} + T_{\text{recovery}}. \quad (7.1)$$

Reported timing statistics include mean, median, standard deviation or IQR, P95/P99 latency, total cycle time, effective cycle time per successful part, and failure or replanning frequency. A conclusion that IL is faster must be based on planning/inference and total cycle-time decomposition, not only on execution time.

7.7 Force-Aware Ablations

The force-aware ablation evaluates RQ4 on both workpieces where feasible. The four variants are A1 pose-only IL, A2 force-observation IL, A3 force-action IL, and A4 proposed force-aware IL. A conventional geometry-based + admittance baseline is included when available. The variants separate observation-side force information, action-side force references, and compliant execution because prior work often combines these components in a single system [5–7,15,22]. Metrics include success rate, desired–measured force RMSE, peak force, tangential force, force variance, contact-loss count, tool slip-out, trajectory tracking error, task time, and unseen-condition performance.

7.8 Evaluation Tables

Table 7.1: Comparison of method capabilities.

Capability	Geometry baseline	Pose-only IL	Force-obs. IL	Force-action IL	Proposed
Explicit coverage	Yes	No	No	No	Local only
Online planning	Yes	No	No	No	No
ROI conditioning	Optional	Yes	Yes	Yes	Yes
Force observation	Optional	No	Yes	No	Yes
Force action	Fixed/profile	No	No	Yes	Yes
Compliant execution	Optional	Optional	Optional	Yes	Yes

Table 7.2: Headlight experiment conditions and metrics.

Study	Conditions	Metrics
ROI mask generalization	Location, size, shape, orientation, curvature region	ROI coverage, spillover, success rate
Local contact execution	Approach, contact, traversal, retract	Force RMSE, variance, peak force, contact loss
Hybrid sequence	Local phase followed by global coverage	feasibility, transition success, total time
Optional real-defect study	Standardized defects only if implemented	defect-area reduction, restoration, over-polishing

Table 7.3: Aluminum pose-randomization conditions.

Variable	Planned randomization	Status
Translation	x, y, z perturbation around anchor poses	[Planned/TBD: TBD ranges]
Orientation	yaw, pitch, and roll perturbation	[Planned/TBD: TBD ranges]
Robot initial configuration	multiple joint-space starts within safety limits	[Planned/TBD: TBD]
Target feature	hole or edge selection	[Planned/TBD: TBD]
Approach direction	nominal and perturbed directions	[Planned/TBD: TBD]
Obstacle or fixture condition	optional condition for collision-aware planning	[Planned/TBD: optional]

Table 7.4: Ablation definitions.

ID	Observation	Action	Execution
A1	vision, robot state	pose trajectory	position or fixed-force control
A2	vision, robot state, F/T history	pose trajectory	position or fixed-force control
A3	vision, robot state	pose + desired force trajectory	admittance or hybrid execution
A4	vision, ROI mask when applicable, robot state, F/T history	pose + desired force trajectory	admittance execution

Table 7.5: Quantitative results template. Values remain TBD until experiments are completed.

Method	Success	ROI/edge quality	Force RMSE	P95 latency	Cycle time
Geometry baseline	TBD	TBD	TBD	TBD	TBD
Rule-based local + global	TBD	TBD	TBD	TBD	TBD
Pose-only IL	TBD	TBD	TBD	TBD	TBD
Force- observation IL	TBD	TBD	TBD	TBD	TBD
Force-action IL	TBD	TBD	TBD	TBD	TBD
Proposed force-aware IL	TBD	TBD	TBD	TBD	TBD

7.9 Statistical Analysis

Each method-condition pair should be repeated across multiple masks, poses, target features, and initial configurations. Method and test split are treated as fixed factors; specimen, anchor pose, and day can be treated as random effects when enough data are available. Timing distributions should be reported with medians, IQR, and high percentiles because planning latency may be heavy-tailed. Failure modes are categorized as perception error, registration error, kinematic infeasibility, collision-planning failure, local-policy perception error, contact instability, mask-generation error, out-of-distribution request, and safety stop.

7.10 Planned and Existing Results

All numerical results in this chapter are planned. The final thesis must distinguish simulated replay, dry-run motion, force-controlled contact without material removal, real deburring or polishing with material removal, and

high-resolution inspection measurements. If the aluminum specimen is already smooth and no burr is measured, the experiment is reported as force-controlled edge-following or deburring-like contact execution rather than deburring-quality validation.

Chapter 8

Conclusions and Future Work

8.1 Conclusions

This thesis draft reformulates the robotic surface-finishing problem as a role-allocation problem between geometric planning and force-aware imitation learning. Geometry-based planning is used where the target surface, tool footprint, and coverage objective can be explicitly specified. ROI-mask-conditioned force-aware IL is used where local behavior depends on the requested region, initial condition, and contact state. A low-level admittance controller provides the shared contact-execution layer.

The proposed framework is evaluated through two planned workpieces. The automotive headlight study tests defect-map- or ROI-mask-conditioned local polishing followed by full-coverage geometric polishing. The ArUco marker is treated only as a temporary command interface for generating an ROI mask and not as a defect-inspection sensor. The aluminum study compares a strong classical perception–registration–planning pipeline with a vision-conditioned IL policy under randomized workpiece pose and robot initial configuration. The key hypothesis is comparable task success with reduced and more consistent online computation time, not task feasibility by IL alone.

The thesis also separates the roles of force observation, learned force reference, and admittance execution. This separation is necessary to show whether force-aware IL improves contact stability beyond pose-only imitation learning and fixed-force execution.

The thesis does not claim novelty from force-aware IL itself, F/T-history-conditioned policies, pose-plus-force prediction, admittance-executed learned policies, mask-conditioned manipulation policies, robotic-polishing IL, learning-geometry integration, or inspection-to-processing workflows. Direct precedents already exist in contact-rich IL, force-policy design, robotic polishing, mask-conditioned manipulation, and industrial automation [1, 5, 7, 14–16, 22, 35].

The actual evaluation target is the task-specific integration: an external defect request represented as an ROI mask, explicit role separation between local interaction-dependent processing and global deterministic coverage, force-aware local pose-force generation, geometry-based full-coverage execution, classical planning versus IL latency/cycle-time comparison, and force observation/action/admittance ablations across the headlight and aluminum workpieces.

8.2 Limitations

This draft does not yet report real-robot quantitative results. Task success, force RMSE, contact-loss count, latency, cycle time, and surface-quality metrics remain [Planned/TBD: to be measured]. The repository also does not contain the final dataset or controller code needed to verify the exact action channel, so the pose-force interface is written with explicit TODOs where implementation details must be confirmed.

The headlight marker interface does not implement autonomous physical-defect inspection. It only emulates an external defect request. The aluminum benchmark does not establish industrial deburring quality unless actual burr removal, residual burr height, chamfer uniformity, and surface damage are measured. The tool-footprint model is approximate and depends on pad compression, curvature, force, and tool wear. The IL policy can fail outside

the demonstrated mask and pose distributions, and force-aware execution still requires conservative safety limits.

8.3 Future Work

Future work includes replacing the ArUco command interface with a real high-resolution defect-inspection system, validating real defect removal and surface restoration, expanding pose and ROI distributions, adding safety-filtered IL or MPC refinement, estimating tool footprint and contact area online, measuring industrial cycle time across larger batches, and integrating material-removal quality feedback. Process-level optimization such as speed, spindle, abrasive, or admittance-gain adaptation should be revisited only after the geometry/IL role allocation and contact-execution layer are validated experimentally.

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Appendix A

Implementation and Experiment Checklists

A.1 Current Implementation Status

- Headlight ROI-mask interface: planned; ArUco ID must be converted to a spatial ROI mask outside the policy.
- Headlight full-coverage planning: planned; tool-footprint model, surface reconstruction, and registration must be verified.
- Aluminum geometry-based baseline: planned; perception, registration, IK, planning, and timing breakdown must be logged separately.
- Force-aware IL policy: architecture defined; exact observation/action channels must be verified from the final dataset and controller implementation.
- Low-level admittance controller: required shared execution layer; selected force/wrench channels and gains must be documented.
- RL velocity optimization and Preston-rate tuning: not a core contribution in the current thesis direction; keep only as optional future work unless separately validated.

A.2 Recommended Dataset Fields

Each demonstration episode should contain timestamps, local RGB or RGB-D images, ROI masks, robot joint state, end-effector pose, linear and angular velocity, measured six-axis force/torque, desired force reference if available, tool or spindle state, action chunk, phase labels, intervention labels, workpiece pose, target feature or ROI metadata, and success or failure annotations.

Marker ID should be stored only as external command metadata and not used as a policy input.

A.3 TODO Items Before Final Submission

- Verify whether the IL action is Cartesian pose, Cartesian increment, joint-space command, normal-force reference, three-axis force reference, or six-axis wrench reference.
- Fill in robot, sensor, controller, tool, fixture, and calibration details.
- Replace figure placeholders with final system, policy, headlight, aluminum, and timing figures.
- Set aluminum randomization ranges and trial counts after safety validation.
- Add quantitative headlight, aluminum, and force-ablation results only after experiments are completed.
- Report unresolved LaTeX warnings, overfull boxes, and citation issues before submission.

A.4 Build Notes

The project is compiled from `ThesisMain.tex`. A typical command is `latexmk -pdf ThesisMain.tex`. The cover can be compiled separately from `ThesisCover.tex`. The author, advisor, submission date, final figures, and experimental results are placeholders that must be updated before submission.

국문요약

적응형 로봇 표면 마감을 위한 비전 기반 기하 경로 계획과 힘 인지 모방학습의 하이브리드 프레임워크

본 학위논문 초안은 로봇 표면 마감 작업에서 기하학적으로 명확히 정의할 수 있는 전면 커버리지와 접촉 상태에 따라 달라지는 국소 상호작용을 분리하여 처리하는 하이브리드 프레임워크를 제안한다. 기하 기반 경로계획은 표면 법선, 경계, 공구 footprint 및 로봇 feasibility를 명시적으로 고려할 수 있으나, 작업물 pose, 로봇 초기 configuration 및 작업 ROI가 반복적으로 변하면 perception, registration, IK 및 motion planning을 매번 다시 수행해야 한다. 반면, 비전 조건 모방학습은 낮고 일정한 online inference latency로 동작을 생성할 수 있으나, 비전만으로는 접촉 압력, 약한 접촉, slip 및 jamming을 충분히 관측하기 어렵고 전면 coverage를 보장하기 어렵다.

제안 방법은 interaction-dependent adaptation이 필요한 국소 작업에는 ROI-mask-conditioned force-aware imitation learning을 사용하고, deterministic coverage가 중요한 영역에는 vision/geometry-based planning을 사용한다. 국소 정책은 RGB 또는 RGB-D observation, 외부에서 생성된 ROI mask, robot proprioceptive state 및 최근 6축 힘/토크 이력을 입력으로 받아 end-effector pose 또는 pose increment와 desired force reference로 구성된 action chunk를 생성한다. 저수준 admittance controller는 측정 wrench와 desired force reference의 오차를 compliant motion correction으로 변환하여 실제 접촉 실행을 담당한다.

두 실증 시편을 중심으로 실험 설계를 구성한다. 자동차 헤드라이트 실험에서는 ArUco marker를 실제 결함 검사 센서로 사용하지 않고, 외부 고해상도 defect-inspection system이 제공한다고 가정한 defect request를 저비용으로 모사하는 command interface로 사용한다. Marker ID는 policy 외부에서 ROI mask로 변환되며, 정책은 marker ID가 아니라 공간 ROI mask만 관측한다. 국소 IL polishing 이후에는 재구성 또는 registration된 headlight surface에 대해 tool-footprint-aware full-coverage polishing을 수행한다. 알루미늄 가공 시편 실험에서는 workpiece pose와 robot initial configuration이 연속적으로

perturbation되는 조건에서 classical perception-registration-planning pipeline 과 vision-conditioned IL policy의 success, online latency, cycle time 및 contact stability를 비교한다.

본 초안은 ROI-mask-conditioned local processing, local-to-global hybrid polishing feasibility, online computation time decomposition, 그리고 F/T observation, learned force action 및 admittance execution의 효과를 검증하는 실험 계획을 제시한다. 실제 작업 성공률, force RMSE, cycle time 및 표면 품질 수치는 아직 보고하지 않으며, 실험 후 [Planned/TBD: 정량 결과 및 표면 검사 지표를 추가해야 한다].

주요어: 로봇 표면 마감, 기하 경로계획, 커버리지 경로계획, 모방학습, 힘/토크 센싱, ROI 마스크, 어드미턴스 제어, 로봇 폴리싱, 로봇 디버링